Mark J. Madler

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Personable and driven Computer Engineering PhD student researching programming models, memory consistency, and verification for RDMA and CXL. Passionate about teaching and engineering design.

Education

University of Colorado Boulder — Boulder, CO
Ph.D., Computer Engineering – Advisor: Dr. Joseph Izraelevitz — *Fall 2024 – Present*
M.S., Computer Engineering — *GPA: 3.97*

Gonzaga University — Spokane, WA
B.S., Computer Engineering — *GPA: 3.66*

Experience

University of Colorado Boulder — August 2024 – Present
Graduate Research Assistant, CU Architecture Lab — *Boulder, CO*

- Research RDMA programming models: encapsulation, memory consistency, and formally verified composable libraries.

Pacific Northwest National Laboratory — May 2025 – March 2026
Research Intern – Mentors: Bruce Palmer, Sayan Ghosh

- Compared one-sided, two-sided, and collective communication for distributed SpMM on NERSC Perlmutter.
- Built PMPI tracing to isolate how inter-process dependencies propagate performance perturbations between ranks.
- Studied the effects of communication models over CXL interconnects.

Kaiser Aluminum — May – December 2023
Automation Engineering Intern

- Implemented industrial automation systems using Ignition SCADA.
- Diagnosed and fixed a process issue causing metal defects using Allen-Bradley PLCs.
- Managed network solutions on a large-scale OT network using vSphere.

Canyon AeroConnect — May – August 2022
Engineering Intern

- Worked on a team to develop a tool for customizing a radio's functional options.
- Developed an internal tool for capturing and filtering telemetry data from radio systems.

Teaching

ECEN 4313/5313: Concurrent Programming — Fall 2026
Teaching Assistant, University of Colorado Boulder

- Built a parallel autograder that grades the commit at the deadline and automatically computes late days.
- Developed in-class demos on lock and atomic performance and false sharing.

Publications

A Verified High-Performance Composable Object Library for Remote Direct Memory Access
G. Ambal, G. Hodgkins, **M. Madler**, G. Chockler, B. Dongol, J. Izraelevitz, A. Raad, V. Vafeiadis — *POPL 2026*

LOCURA: Lazy Consistency for RDMA Distributed Shared Memory
M. Madler, G. Ambal, E. Karabacak, B. Palmer, S. Ghosh, J. Izraelevitz — *EuroSys 2027 (in prep.)*

- RDMA DSM using lazy release consistency and multi-writer cachelines; formally proven correct.
- 1.9× faster than MPI RMA on Graph500; 3–139× faster than Argo DSM and 1.2–6× faster than UPC++.

Understanding Performance Impacts of Communication Models on Sparse Matrix Multiplication
M. Madler, B. Palmer, S. Ghosh, J. Izraelevitz — *IEEE CAL (in prep.)*

- Shows one-sided communication wins by decoupling inter-process dependencies, not by curing load imbalance.

FLAME: Function-Level Analysis and Merging with Embeddings
H. AlGhanim, **M. Madler**, H. Malahieude, J. Izraelevitz — *CGO 2027 (under review)*

- Embedding-guided LLVM function merging: 11.02% size reduction over `-Oz` vs. 9.06% for F3M.

Skills

Languages:	C, C++, Python, C#, Assembly, MATLAB, VHDL, Verilog
HPC & Systems:	RDMA, MPI RMA, UPC++, Global Arrays, CXL, pthreads, LLVM, eBPF, gem5
Tools:	Linux, Git/GitHub, Docker, Make/CMake, LaTeX, Slurm, vSphere, Ignition, SolidWorks